

14.4 Tangent Planes and Normal Lines

1 Warm Up

Problem 1: Find the plane going through $P(-1, 1, 2)$ normal to $2\hat{i} + 3\hat{j} - \hat{k}$.

Problem 2: Finds the line going through $P(-1, 1, 2)$ in the direction of $2\hat{i} + 3\hat{j} - \hat{k}$.

2 Motivation

Now that we know how to take the derivative at ANY direction not just x and y , we can ask other interesting questions in 3D that we don't have in 2D. For example we can now have a tangent *plane* to a surface at a point. Here we utilize what we have learned to find the equation for a tangent plane along with a normal line!

3 Definitions & Theorems

1. In order to find the tangent plane to a point on a surface $z = f(x, y)$ it boils down to finding the normal vector to the plane. We know that any constant plane $z = c$ will intersect the surface at a level curve which we project onto the xy plane (2-dimensional).
2. Since all continuous curves can be represented by a vector function, we get that our level curve can be written as,

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j}$$

3. For any level curve at $z = c$, we see that $f(x(t), y(t)) = c$. Taking the derivative with respect to t of both sides, we get, $f'(x(t), y(t)) = 0$ and then using the chain rule we see,

$$\begin{aligned}\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} &= 0 \\ \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \right) \cdot \left(\frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} \right) &= 0 \\ \nabla f(x, y) \cdot \vec{r}'(t) &= 0\end{aligned}$$

4. When is the dot product zero? The vectors are **orthogonal**. Since $r'(t)$ gives me the slope of the tangent vector to the level curve, the gradient must give us the **normal** to any point on a level curve. (Also the steepest climb which can be seen with contour maps).
5. If $\nabla f(x, y)$ is the normal to the level curve $f(x, y) = c$, then $\nabla f(x, y, z)$ is the normal to the level surface $f(x, y, z) = c$.
6. So in order to find the normal to a surface, we can find the normal to a level curve just one dimension higher. For example if we have the 2-D curve $x^2 + y^2 = 16$ and we want the normal to this curve we need to take the gradient of a surface, i.e. $f(x, y) = x^2 - y^2$.

4 Examples

Problem 1. Find the tangent plane and the normal line to surface $xyz = -4$ at the point $P(2, -1, 2)$

Problem 2. Find the tangent plane and the normal line to the surface $xz^2 + yx^2 + y^2 - 2x + 3y + 6 = 0$ at the point $P(-2, 1, 3)$.

Problem 3. Find the tangent plane and the normal line to the surface $z = \ln(x/y)$ at the point $P(e, 1, 1)$.

Problem 4. Find the tangent plane and the normal line to the surface $z = \tan^{-1}(y/x)$ at the point $P(1, 1, \pi/4)$.

5 Scratch Work